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The severity of human peri-implantitis lesions correlates with the level of submucosal microbial dysbiosis

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Abstract

Aim: To cross-sectionally analyse the submucosal microbiome of peri-implantitis (PI) lesions at different severity levels.

Materials and Methods: Microbial signatures of 45 submucosal plaque samples from untreated PI lesions obtained from 30 non-smoking, systemically healthy subjects were assessed by 16s sequencing. Linear mixed models were used to identify taxa with differential abundance by probing depth, after correction for age, gender, and multiple samples per subject. Network analyses were performed to identify groups of taxa with mutual occurrence or exclusion. Subsequently, the effects of peri-implant probing depth on submucosal microbial dysbiosis were calculated using the microbial dysbiosis index.

Results: In total, we identified 337 different taxa in the submucosal microbiome of PI. Total abundance of 12 taxa correlated significantly with increasing probing depth; a significant relationship with lower probing depth was found for 16 taxa. Network analysis identified two mutually exclusive complexes associated with shallow

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pockets and deeper pockets, respectively. Deeper peri-implant pockets were associated with significantly increased dysbiosis.

Conclusion: Increases in peri-implant pocket depth are associated with substantial changes in the submucosal microbiome and increasing levels of dysbiosis.

KEYWORDS

16s, dysbiosis, microbiome, next-generation sequencing, peri-implant disease, peri-implantitis

1 | INTRODUCTION

Peri-implantitis (PI) is an increasingly prevalent chronic inflammatory disease, with up to 22% of patients being affected (Derks & Tomasi, 2015). Similar to periodontitis, the chronic inflammatory condition affecting teeth caused by dysbiotic biofilms in susceptible hosts, principal clinical signs of peri-implantitis include pocketing, bleeding on probing, and/or suppuration. Both diseases lead to the destruction of supporting tissues and, in terminal stages, to the loss of the implant or the tooth (Lindhe & Meyle, 2008). Both diseases share common risk factors, including environmental exposures, such as smoking and poor oral hygiene, and systemic predispositions such as diabetes. Interestingly, a history of periodontitis has been shown to increase the risk of subsequent PI, suggesting commonalities in their aetiology (Renvert & Quirynen, 2015; Sousa et al., 2016; Stacchi et al., 2016). Still, PI differs from periodontitis with respect to anatomic features and cellular composition of the lesion (Berglundh, Zitzmann, & Donati, 2011; Carcuac & Berglundh, 2014) and, most notably, to disease progression rate which is significantly higher in PI than in periodontitis (Salvi, Cosgarea, & Sculean, 2017).

Thus far, the pathophysiological underpinnings accounting for the faster progression rate in PI are poorly understood. Distinctions in key features of the lesion-associated microbiomes of the two diseases can conceivably play a role in this respect. Specifically, PI lesions of increasing clinical severity may be associated with specific microbial signatures that may result in accelerated progression.

Thus far, no studies have investigated the relationship between PI severity and the associated submucosal microbial patterns. Previous work has mainly focused on identifying microbial differences between healthy and diseased peri-implant sites (Al-Ahmad et al., 2018; Apatzidou et al., 2017; Koyanagi, Sakamoto, Takeuchi, Ohkuma, & Izumi, 2010; Kumar, Mason, Brooker, & O'Brien, 2012; Maruyama et al., 2014; Sanz-Martin et al., 2017; Shiba et al., 2016; Tamura, Ochi, Miyakawa, & Nakazawa, 2013; Zheng et al., 2015). Importantly, confounders including general health status, smoking, age, gender, and prior treatment have generally not been accounted for in the existing literature, hampering inferences on the microbial features of PI at different levels of disease severity.

In this work, we therefore sought to characterize the submucosal microbiome of PI lesions at different severity levels. To do so, we used 16s rRNA sequencing to analyse the previously untreated PI lesions from systemically healthy, non-smoking subjects. We identified taxa with significant associations with deeper or shallower peri-implant pockets and demonstrate a link between deeper probing depths and increasing levels of microbial dysbiosis.

Clinical Relevance

Scientific rationale for the study: Peri-implantitis (PI) is increasingly prevalent and poses a significant clinical challenge. Thus far, the microbial signatures of PI sites of different severity have not been characterized. Specifically, it is unknown whether deeper peri-implant pockets feature an increased level of dysbiosis.

Principle findings: The microbiome of PI lesions consists of both common periodontal bacteria and novel species. Peri-implant probing depth correlates with distinct microbial profiles, and deeper pockets show higher levels of dysbiosis.

Practical implications: Peri-implant pockets contain disease-associated biofilms that become increasingly dysbiotic with increasing probing depth.

2 | MATERIALS AND METHODS

2.1 | Study population and clinical procedures

The protocol for the study was approved by the institutional review boards/ethics committees at the Universities of Bonn, Würzburg, Tübingen and Düsseldorf, Germany, and the regional ethics committee of the Jönköping district, Sweden. The study was conducted in accordance with the guidelines of the World Medical Association Declaration of Helsinki. Each participant was informed of potential risks and benefits of this study and gave written informed consent.

Study participants were recruited at the dental clinics of the universities of Bonn, Würzburg, Tübingen, Düsseldorf and the Dental Hospital Jönköping. Participants had to be at least 18 years old, non-diabetic, non-smokers, or former smokers (>6 months) who did not receive any antibiotics or anti-inflammatory drugs within the past 6 months prior to the examination. Subjects that qualified for the

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TABLE 1 List of selected study characteristics

Study characteristics	
Population-specific	30 patients
Female	70% (21 individuals)
Male	30% (nine individuals)
Mean age	64.7 ± 11.5 years (range: 32–82 years)
Mean number of teeth	15.85 ± 8.04 (range: 0–27 teeth)
Mean number of dental implants	3.46 ± 1.6 (range: 1 to ≥8 implants)
Site-specific	45 implants with peri-implantitis Inclusion criteria: • Implant sites qualified for CIST (Lang et al., 2000) therapy scheme D • PD ≥5 mm • BoP + • Bone loss ≥3 mm
Mean PDmax	8.00 ± 2.04 mm (range: 5–11 mm)
Suppuration	Observed in 73% of sites

diagnosis of periodontitis as a manifestation of systemic diseases and subjects that were pregnant or lactating were excluded. All participants had at least one dental implant with PI that had not received any therapeutic intervention in the preceding 6 months, except for occasional supragingival prophylaxis. Sites had to qualify for cumulative interceptive supportive therapy C.I.S.T., (Lang, Wilson, & Corbet, 2000) scheme grade D; that is, probing depth (PD) ≥5 mm, the presence of bleeding on probing (BoP), and bone loss ≥3 mm. All participants underwent a full-mouth periodontal examination, and each implant was probed at six sites and the maximum PD (PDMax). BoP/suppuration, implant/surface type, reconstruction method, and the width of keratinized mucosa were recorded.

2.2 | Sample harvesting and sequencing preparation

Submucosal plaque samples were obtained from each PI-affected implant before surgical intervention, using one sterile paper point at each of the mesial, distal, buccal, and lingual aspects of the implant, and were subsequently pooled (Jervoe-Storm, Alahdab, Koltzscher, Fimmers, & Jepsen, 2007). Sample tubes were stored immediately at −20°C or −80°C until further processing. DNA was extracted and purified using spin columns (Sigma-Aldrich GenElute Bacterial Genomic DNA Kit, Sigma-Aldrich, Munich, Germany). Quality and quantity of DNA was assessed spectrophotometrically using a NanoDrop spectrophotometer.

2.3 | Sequence processing

Sequencing data of the V3-V4 hypervariable region of the 16s rRNA gene were obtained using the Illumina MiSeq platform (Illumina, San Diego, California, USA) and a paired-end approach with 2 × 300 bp reads (Caporaso et al., 2011) providing substantial depth and breadth of coverage (Frey et al., 2014; Nelson, Morrison, Benjamino, Grim, & Graf, 2014). Library preparation was performed following the Illumina 16S rRNA Sequencing library preparation guide (Illumina, 2013) using the Nextera XT Index Kit (Illumina), PhiX Control Kit (Illumina), and MiSeq V3 reagent kit (Illumina). Initial purity and quantity was measured on a NanoDrop spectrophotometer. Subsequently, a PCR amplification was performed using the KAPA HiFi Hotstart ReadyMix (KAPABIOSYSTEMS, Boston, Massachusetts, USA) and the primer pair 5′-TCGTCGGCAGCGTCAGATGTGTATAAGAGACAGCCTACGGGNGGCWGCAG and 5′-GTCTCGTGGGCTCGGAGATGTGTATAAGAGACAGGACTACHVGGGTATCTAATCC (Klindworth et al., 2013). Sequencing of the obtained libraries followed the manufacturer's instructions. The mean paired-end reads/sample were 109,945 ± 35,169.

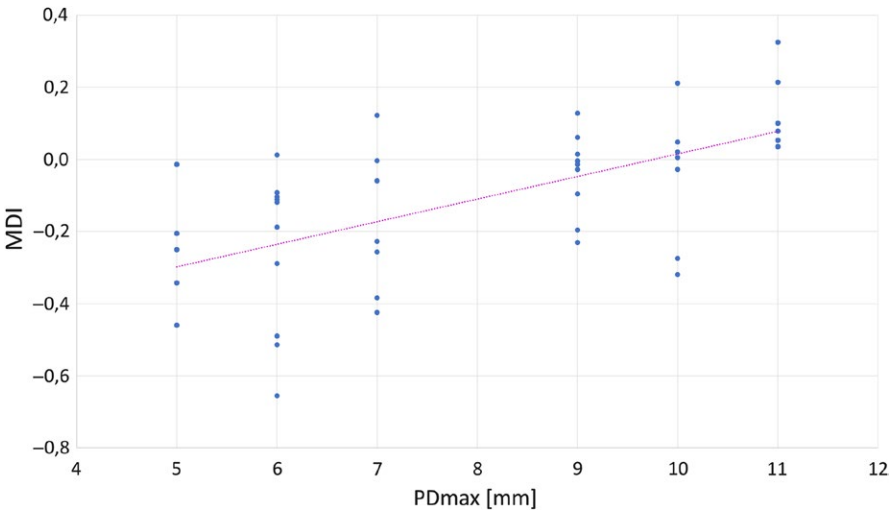

FIGURE 1 Microbial Dysbiosis Index (MDI) of each sample plotted against probing depth (PDmax [mm]). The slope of the line is statistically significant ($p = 1.32 \times 10^{-7}$). MDI increases with growing PDmax, indicating gain of taxa abundance associated with disease severity, whilst taxa with opposite association show decreased abundance

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Quality control of the obtained reads was performed using FastQC (Andrews, 2010). Data processing, including demultiplexing, quality filtering to a minimum Phred score of 20, and joining of paired-end reads, was carried out using QIIME 1.9.1 (Caporaso, Kuczynski, et al., 2010). In the next step, the open reference picking strategy was selected. Sequences were clustered with a similarity threshold of 97% via UCLUST (Edgar, 2010) to operating taxonomic units (OTUs). Assignment of representative sequences, picked by QIIME, was done by the RDP classifier v2.2 (Wang, Garrity, Tiedje, & Cole, 2007), and alignment to the reference database, the Human Oral Microbiome Database v15.1 (Chen et al., 2010) was conducted via PyNAST (Caporaso, Bittinger, et al., 2010). OTU data were normalized using the DESeq2 approach in QIIME (Love, Huber, & Anders, 2014; McMurdie & Holmes, 2014). α-diversity through whole tree PD (Moritz & Faith, 1998), a measure of abundance-based richness, was assessed using QIIME. The same platform was used to assess β-diversity by means of unweighted UniFrac (Lozupone & Knight, 2005).

2.4 | Statistical analysis

The relationship of the deepest PD per implant (PDMax) with the abundance of each OTU was assessed through regression analysis using linear mixed models in R 3.3.3 (R Development Core Team, 2008), utilizing the lme4 1.1-15 package (Bates, Mächler, Bolker, & Walker, 2015) in conjunction with lmerTest 2.0-36 to test for significance (Kuznetsova, Brockhoff, & Christensen, 2017). The higher within-subject correlation was taken into account by modelling the subjects as a random effect. Furthermore, gender and age distribution were added to the model as fixed factors. Results were corrected for multiple testing through FDR adjustment (Benjamini, Drai, Elmer, Kafkafi, & Golani, 2001).

Taxa significantly associated with probing depth were considered in the network analysis. Taxa co-occurrence and/or co-exclusion were determined via renormalization and permutation with the R package ccrepe ("Compositionality Corrected by Renormalization and

TABLE 2 Abundance of taxa associated with PDmax. Positive estimates stand for positive correlation and vice versa. Only taxa with an FDR-corrected *p*-value <0.05 are shown. Residual taxa and analyses on other taxa level can be viewed in Supporting Information Table S2

Taxon	Estimate	Std.-error	*p*-value	FDR-corr.
Eubacteriaceae [XV]	0.7598	0.1414	0.0000	0.0014
Fretibacterium sp. *HMT 362*	0.4660	0.0938	0.0001	0.0088
Selenomonas sp. *HMT 136*	−0.7589	0.1695	0.0001	0.0095
Granulicatella elegans	−0.6547	0.1450	0.0002	0.0135
Fretibacterium fastidiosum	0.4069	0.0916	0.0002	0.0135
Rothia aeria	−0.7634	0.1916	0.0003	0.0167
Corynebacterium durum	−0.3884	0.1046	0.0006	0.0206
Peptostreptococcaceae [XI][G-6]	0.5954	0.1561	0.0005	0.0206
Veillonella dispar	−0.5955	0.1539	0.0005	0.0206
Acinetobacter	−0.6184	0.1526	0.0006	0.0206
Streptococcus oralis subsp. *dentisani clade 398*	−0.4994	0.1313	0.0009	0.0267
Alloprevotella sp. *HMT 473*	0.4511	0.1222	0.0010	0.0286
Veillonella parvula	−0.4540	0.1233	0.0012	0.0310
Fastidiosipila sanguinis	0.3670	0.1065	0.0014	0.0326
Bifidobacterium	−0.5306	0.1563	0.0019	0.0332
Bergeyella	−0.4549	0.1326	0.0018	0.0332
Filifactor alocis	0.4915	0.1312	0.0017	0.0332
Veillonella rogosae	−0.4570	0.1367	0.0018	0.0332
Neisseria	−0.6571	0.1906	0.0016	0.0332
Veillonella atypica	−0.6161	0.1904	0.0025	0.0414
Rhodobacteraceae	−0.4294	0.1328	0.0026	0.0416
Peptostreptococcaceae [XI][G-4]	0.4873	0.1424	0.0028	0.0424
Bacteroidetes [G-3] bacterium HMT 365	0.4473	0.1322	0.0034	0.0468
Lautropia mirabilis	−0.5693	0.1793	0.0032	0.0468
Treponema parvum	0.5249	0.1628	0.0035	0.0468
Clostridiales [F-1][G-1] bacterium HMT 093	0.4993	0.1516	0.0037	0.0473
Oribacterium	0.4296	0.1377	0.0041	0.0496
Oribacterium asaccharolyticum	−0.3888	0.1272	0.0040	0.0496

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Permutation") 1.10.0, designed to detect correlations in microbial data using the nc.score as the similarity measure.

To assess the extent of dysbiosis of the biofilm associated with each implant site, we utilized the Microbial Dysbiosis Index (MD-index; Gevers et al., 2014), calculated as follows:

$$\mathrm{MDI} = \log\left(\frac{[\text{total abundance of OTUs positively associated with PDmax}]}{[\text{total abundance of OTUs negatively associated with PDmax}]}\right)$$

Regression analysis was run with mixed effects linear models correcting for gender, age, and within-subject correlation to detect possible associations of Microbial Dysbiosis Index (MDI) and PD, as outlined above.

Analysis of microbiome characteristics of deep and shallow pocket groups was conducted via chi-square-test (Chernoff & Lehmann, 1954).

3 | RESULTS

3.1 | Study population

A total of 45 implants with PI, obtained from 30 subjects, were included in the present analyses. As shown in Table 1, 70.0% of the participants were female with a mean age of 64.7 ± 11.5 years (range 32–82 years). On average, participants had 15.85 ± 8.04 teeth and 3.46 ± 1.61 dental implants. Mean PDmax was 8.00 ± 2.04 mm (range 5–11 mm). All implants showed BoP, and suppuration was observed at 73% of the sites.

3.2 | Structure and diversity of the bacterial community

All samples provided sufficient DNA quality and quantity for analysis. OTUs were classified in 14 phyla, 169 genera, and 337 species-level phylotypes. A complete list of observed taxonomic units can be viewed in the Supporting Information Table S1.

3.3 | Dysbiosis

A highly significant ($p = 1.32 \times 10^{-7}$) positive association of the MD-index with increasing PDmax was observed (Figure 1).

3.4 | Characteristics of the microbiome in relation to disease severity

To characterize the microbial shifts in relation to PDmax leading to the observed increase in dysbiosis, we performed linear mixed model analyses of the normalized relative abundance of observed phylotypes on the species-level in relation to PDmax. We identified 28 significantly differentially abundant taxa after correction for age and gender (Table 2). Abundance of the taxa *Eubacteriaceae [XV]*, *Fretibacterium* sp. *HMT 362*, *Fretibacterium fastidiosum*, *Peptostreptococcaceae [XI][G-6]*, *Alloprevotella* sp. *HMT 473*, *Fastidiosipila sanguinis*, *Filifactor alocis*, *Peptostreptococcaceae [XI][G-4]*, *Bacteriodetes [G-3] bacterium HMT 365*, *Treponema parvum*, *Clostridiales [F-1][G-1] bacterium HMT 093*, and *Orobacterium* was significantly (FDR-corrected $p < 0.05$) higher with increasing PDmax, whereas the taxa *Selemonas* sp. *HMT 136*, *Granulicatella elegans*, *Rothia aeria*, *Corynebacterium durum*, *Veillonella dispar*, *Acinetobacter*,

TABLE 3 Characteristics of the taxa significantly associated with PDmax

Taxa	Gram +/−	Anaerobic/ aerobic
Positive association with PDmax		
Eubacteriaceae [XV] (Ludwig, Schleifer, & Whitman, 2009)	+	NA
Fretibacterium sp. *HMT 362*[a]	−	Anaerobic
Fretibacterium fastidiosum[a]	−	Anaerobic
Peptostreptococcaceae [XI][G-6][b]	+	Anaerobic
Bacteroidetes [G-3] bacterium HMT 365[a]	NA	Anaerobic
Peptostreptococcaceae [XI][G-4][b]	+	Anaerobic
Oribacterium[a]	NA	Anaerobic
Treponema parvum[a]	−	Anaerobic
Filifactor alocis (Jalava & Eerola, 1999)	+	Anaerobic
Alloprevotella sp. *HMT 473*[a]	NA	Anaerobic
Fastidiosipila sanguinis[a]	+	Anaerobic
Clostridiales [F-1][G-1] bacterium HMT 093 (Yutin & Galperin, 2013)	+	Anaerobic
Negative association with PDmax		
Bifidobacterium (Schell et al., 2002)	+	Anaerobic
Selenomonas sp. *HMT 136*[a]	−	Anaerobic
Granulicatella elegans[a]	+	Anaerobic
Rothia aeria[a]	+	Aerobic
Acinetobacter (Bitrian, Gonzalez, Paris, Hellingwerf, & Nudel, 2013)	−	Aerobic
Streptococcus oralis subsp. *dentisani clade 398*[a]	+	Anaerobic
Neisseria[a]	−	Aerobic
Veillonella parvula[a]	−	Anaerobic
Rhodobacteraceae (Simon et al., 2017)	−	NA
Bergeyella (Shukla et al., 2004)	−	Aerobic
Veillonella rogosae[a]	−	Anaerobic
Lautropia mirabilis[a]	−	Anaerobic
Veillonella atypica[a]	−	Anaerobic
Veillonella dispar[a]	−	Anaerobic
Oribacterium asaccharolyticum[a]	+	Anaerobic
Corynebacterium durum[a]	+	Aerobic

[a]Information was compiled from BacDive (Sohngen et al., 2016). [b]Information was compiled from the American Type Culture Collection database.

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Streptococcus oralis subsp. *dentisani clade 398*, *Veillonella parvula*, *Bifidobacterium*, *Bergeyella*, *Veillonella rogosae*, *Neisseria*, *Veillonella atypica*, *Rhodobacteriaceae*, *Lautropia mirabilis*, and *Oribacterium asaccacrolyticum* were significantly (FDR-corrected $p < 0.05$) decreased in samples with increasing PDmax. Virtually identical results were obtained when using alternative methodology. Results on different taxonomic levels can be found in the Supporting Information Table S2.

Taxa associated with increasing PDmax were exclusively anaerobes, whilst taxa associated with lower PDmax showed significantly different ($p < 0.03$), more variable oxygen requirements (Table 3). α-diversity was significantly ($p < 0.033$) decreased in groups with deeper pockets (>7 mm; Figure 2) demonstrating higher species richness in samples from shallower pockets (≤7 mm) and vice versa. Collectively, the principle coordinate analysis plots PCoA (Figure 3) suggest that the second principle coordinate (PC2) layers the samples by PDmax (A), which is positively associated with MD-index (C), representing similar species diversity of samples with similar probing depths. α-diversity, on the other hand, tends to be distributed on PC1. The scatter density plot of the samples of whole tree PD versus MD-index suggests that samples with higher MD-index (>0.1) are characterized by markedly reduced species richness, as compared to the samples with lower MD-index.

3.5 | Correlation network

Correlation analysis using the ccrepe package showed a clear division into two groups of cooperation units. *L. mirabilis*, *Neisseria*, *G. elegans*, *Bergeyella*, *R. aeria*, *C. durum*, *Oribacterium asaccharolyticum*, *Selemonas* sp. *HMT 136*, *V. rogosae*, *V. parvula*, *V. atypica*, *V. dispar*, *Streptococcus oralis* subsp. *dentisani clade 398*, and *Bifidobacterium* formed one complex with strong co-occurrence of taxa that negatively associated with PDmax. A second, smaller complex negatively associated with PDmax was formed by the taxa *Acinetobacter* and *Rhodobacteraceae*. These two complexes showed mutual exclusion with the three complexes comprised of taxa positively associated with PDmax: (a) *F. alocis*, *Bacteroidetes bacterium HMT 365*, and *Clostridiales [F-1][G-1] bacterium HMT 093*; (b) *Peptostreptococcaceae [XI][G-6]*, *Peptostreptococcaceae [XI][G-4]*, *Fretibacterium* sp. *HMT 362*, and *F. fastidiosum;* and (c) *Eubacteriaceae [XV]*. Figure 4 shows a visualization of the network, detailed results are captured in Table 4. Additionally, findings from genus-level analysis are presented in the Supporting Information Table S4 and Figure S1.

4 | DISCUSSION

Our findings reveal significant associations of specific features of the submucosal microbiome from PI lesions with PI severity, expressed through the depth of the peri-implant pocket. Our analyses cover multiple levels of the submucosa bacterial ecology and aspects of co-occurrence and co-exclusion. To the best of our knowledge, this is the first study investigating these associations on a whole-microbiome level, and the first study that included microbial samples from a well-defined patient cohort specifically enrolled to minimize bias introduced by subject-related factors (including smoking, diabetes, or medications) and local factors (such as prior therapeutic interventions at the site). Possible remaining confounders were addressed using multi-variate analyses.

In analyses examining the effects of ecological shifts on disease severity, we observed that the MD-index, a measure of submucosal bacterial dysbiosis, is positively associated with increasing peri-implant pocket depths, suggesting that potential pathogens outcompete health-associated taxa (Marsh, 2015), similar to earlier observations derived from several other human diseases—including diabetes, autism, obesity, and periodontitis (Boulange, Neves, Chilloux, Nicholson, & Dumas, 2016; Buffington et al., 2016;

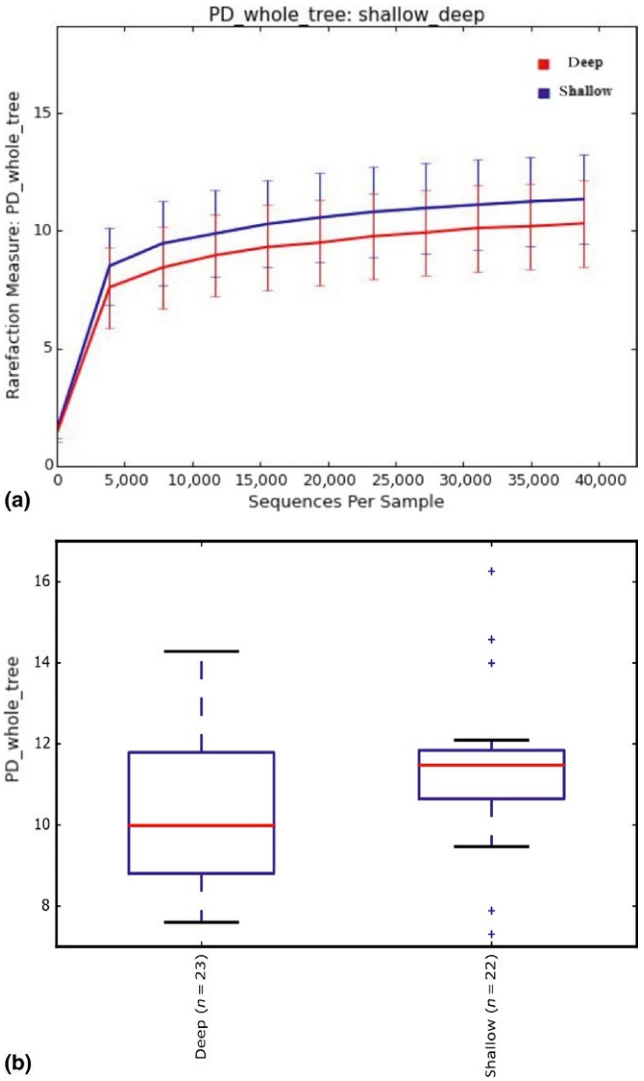

FIGURE 2 Results of α-diversity analysis: (a) Rarefaction plot: samples with shallow pockets (blue) show larger whole tree probing depth (PD) measures. Samples of deeper pockets (PDmax >7 mm) on the other hand demonstrate lower species richness. (b) Box plot diagram of whole tree PD measure of shallower and deeper pocket sample groups. α-diversity is significantly ($p < 0.033$) decreased in latter group

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Carding, Verbeke, Vipond, Corfe, & Owen, 2015). To the best of our knowledge, our study is the first to show such an association in the context of human PI.

When analysing the microbial profiles underlying the observed dysbiosis, our findings indicate that PI lesions are primarily inhabited by anaerobic Gram-negative taxa, corroborating previous studies by other investigators (Al-Ahmad et al., 2018; Apatzidou et al., 2017; Koyanagi et al., 2010; Kumar et al., 2012; Sanz-Martin et al., 2017; Tamura et al., 2013). All aforementioned studies compared peri-implant health and PI, whereas our study focused on PI lesions with differing disease severity. Interestingly, the taxa identified in our analyses as associated with shallower peri-implant pocket depth do overlap with those identified as associated with health in case–control studies, for example, *Streptococcus* and *Veillonella*.

In addition, we identified several taxa not previously reported to occur at PI sites, including *O. asaccharolyticum*, *V. dispar*, *G. elegans*, and *C. durum*, that is, natural inhabitants of the human aerodigestive tract (Chen et al., 2018; Kumar et al., 2006; Ohara-Nemoto et al., 2005; Sizova et al., 2014).

A number of the taxa identified to be associated with deeper peri-implant pockets have been associated with severe periodontal disease, for example, *F. alocis* (Aruni, Chioma, & Fletcher, 2014; Aruni et al., 2015), *F. fastidiosum* (Correa et al., 2017; Moon, Lee, & Lee, 2015; Vartoukian, Downes, Palmer, & Wade, 2013), *Alloprevotella*, *Clostridiales* (Chen et al., 2018), or *Treponema* (Socransky, Haffajee, Cugini, Smith, & Kent, 1998), especially *T. parvum*, which was also found in sites originating from acute necrotizing ulcerative gingivitis lesions (Wyss et al., 2001). In addition, we also identified significant positive associations with taxa not previously linked to PI or periodontitis, for example, *F. sanguinis*, a species recently discovered in human blood samples (Falsen et al., 2005) but hitherto not related to human nosology.

In addition, the dysbiotic shift observed in the transition from shallower to deeper pockets is characterized by decreasing species richness: The group consisting of exclusively anaerobic, mostly gram-negative taxa, some of which are known periodontal pathogens, outcompeted the health-associated taxa that were prevalent in samples with shallower probing depths.

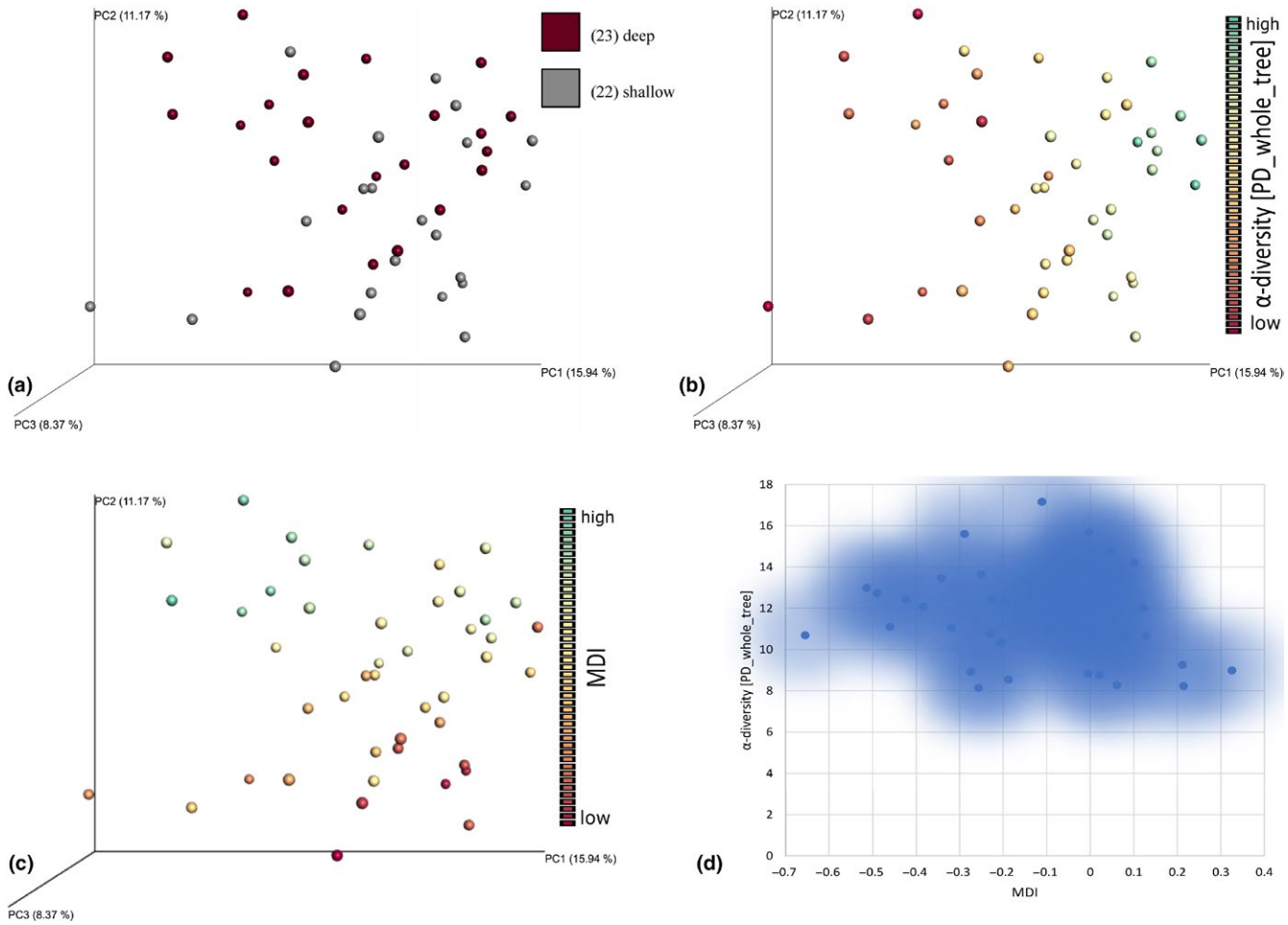

FIGURE 3 Principle coordinate plots, whereas each axis exemplifies one of the three top principle coordinates seizing the major part of the diversity. Shown in per cent, the fraction of diversity is described. (a) shows the samples coloured by grouping of deeper (probing depth [PD]max >7 mm) and shallower pockets (PDmax ≤7 mm), (b) by the scope of whole tree PD, and (c) reflecting the gradient of Mean Dysbiosis Index (MDI). The scatter plot in (d) of alpha diversity versus MDI: samples with increased MDI (>0.1) show a sudden reduction in species richness

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Network analysis facilitated the study of inter-related taxa and further identified two groups of mutually exclusive complexes. Interestingly, we found the taxa in the first, larger group to be associated with shallow pockets, whilst the taxa in the second group were significantly more numerous in deeper pockets. These findings indicate that PI is clearly not a mono-infection by a specific pathogen, but rather a multifactorial, dysbiosis-associated process. Specifically, within the limits of a study only including "diseased" implants, our data point to the importance of a balanced peri-implant ecosystem, with an equilibrium between health-associated bacteria and pathogens (Socransky & Haffajee, 2005).

Strengths of our study include stringent inclusion criteria, a sizeable patient sample, and use multivariate analyses to account for possible residual confounders. Limitations include the cross-sectional nature of the study and the absence of a control group of implants without PI that obviously preclude inferences regarding

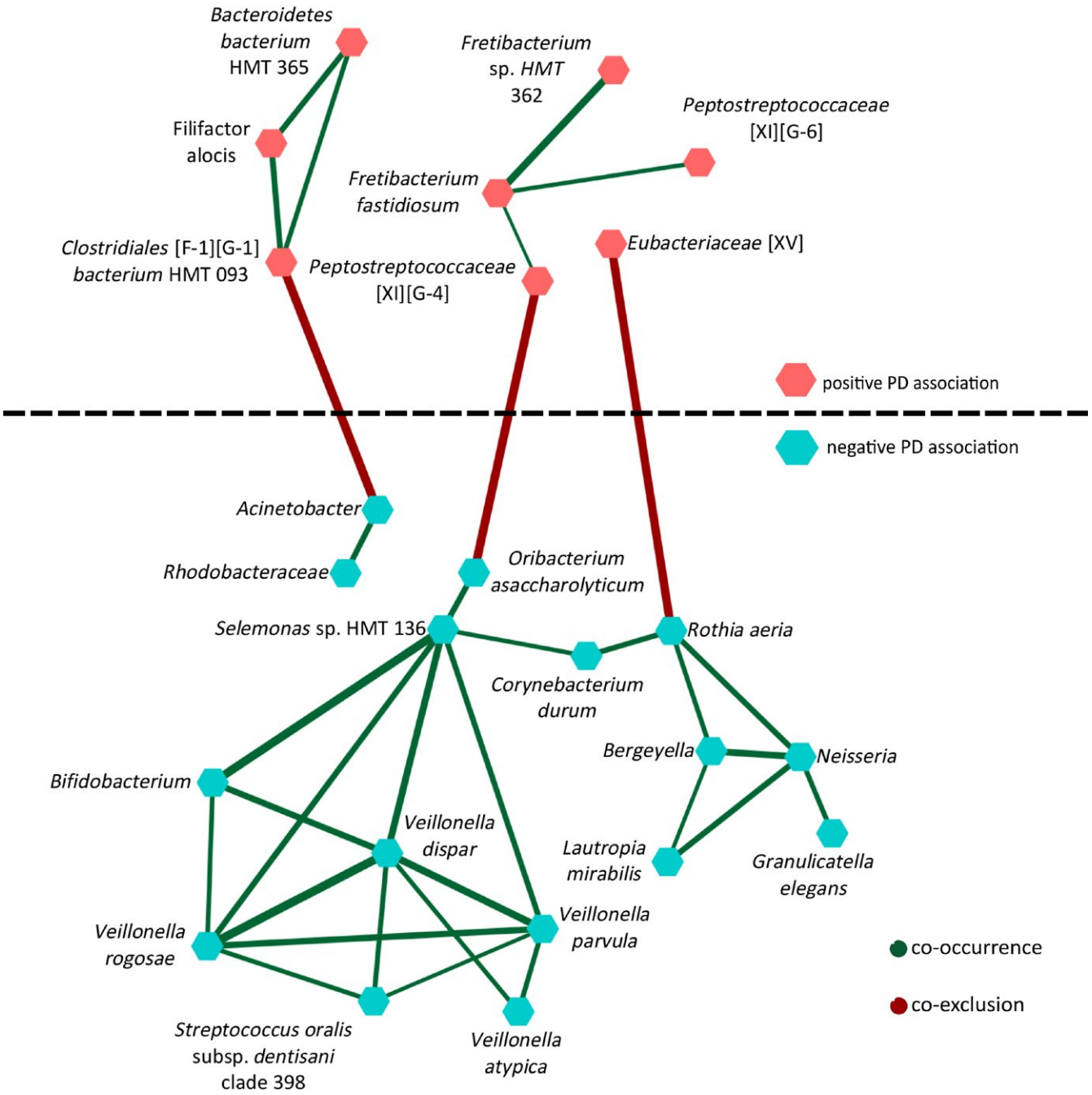

FIGURE 4 Network of taxa significantly associated with disease severity. Red lines indicate co-exclusion whilst green lines indicate co-occurrence. The results were filtered for better visualization: Correlations with FDR-corrected *p*-values <0.1. Weights vary between −0.424 and 0.673. The width of the lines mirrors the weight strengths. Taxa with positive association with pocket depth appear above the broken line without co-occurring taxa associated with shallower pockets which cluster below the broken line. Results can be viewed in Table 2 (unfiltered version in Supporting Information Table S3)

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TABLE 4 Edgelist of the network analysis. It displays interaction of two taxa, whereas each line represents one single interaction. Negative values in the weight column mean co-exclusion, whilst positive values indicate co-occurrences. Only edges with significant FDR-corrected *p*-values <0.1. The complete list can be displayed in the Supporting Information Table S3

From	To	Weight	FDR-corr.	Interaction type
Selemonas sp. *HMT 136*	*Bifidobacterium*	0.6610	0.0000	Co-occurrence
Fretibacterium sp. *HMT 362*	*Fretibacterium fastidiosum*	0.5888	0.0000	Co-occurrence
Veillonella rogosae	*Veillonella dispar*	0.6734	0.0000	Co-occurrence
V. dispar	*Selemonas* sp. *HMT 136*	0.5609	0.0000	Co-occurrence
Neisseria	*Bergeyella*	0.5491	0.0000	Co-occurrence
Veillonella parvula	*V. dispar*	0.5849	0.0000	Co-occurrence
V. dispar	*Bifidobacterium*	0.5262	0.0001	Co-occurrence
Acinetobacter	*Rhodobacteraceae*	0.4874	0.0002	Co-occurrence
Neisseria	*Rothia aeria*	0.4863	0.0003	Co-occurrence
Neisseria	*Lautropia mirabilis*	0.4999	0.0003	Co-occurrence
V. rogosae	*V. parvula*	0.5269	0.0014	Co-occurrence
V. dispar	*Streptococcus oralis* subsp. *dentisani clade 398*	0.4422	0.0036	Co-occurrence
Neisseria	*Granulicatella elegans*	0.4429	0.0037	Co-occurrence
Selemonas sp. *HMT 136*	*Oribacterium asaccharolyticum*	0.4953	0.0038	Co-occurrence
Corynebacterium durum	*R. aeria*	0.4692	0.0047	Co-occurrence
V. parvula	*Selemonas* sp. *HMT 136*	0.4817	0.0055	Co-occurrence
V. rogosae	*Selemonas* sp. *HMT 136*	0.5026	0.0059	Co-occurrence
V. dispar	*Veillonella atypica*	0.4132	0.0065	Co-occurrence
Filifactor alocis	*Bacteroidetes bacterium HMT 365*	0.4879	0.0073	Co-occurrence
Eubacteriaceae [XV]	*R. aeria*	−0.4064	0.0082	Co-exclusion
V. rogosae	*Streptococcus oralis* subsp. *dentisani clade 398*	0.4195	0.0106	Co-occurrence
F. alocis	*Clostridiales [F-1][G-1] bacterium HMT 093*	0.4911	0.0110	Co-occurrence
Bergeyella	*R. aeria*	0.4292	0.0114	Co-occurrence
Peptostreptococcaceae [XI] [G-4]	*O. asaccharolyticum*	−0.4226	0.0208	Co-exclusion
L. mirabilis	*Bergeyella*	0.3834	0.0423	Co-occurrence
F. fastidiosum	*Peptostreptococcaceae [XI][G-6]*	0.4005	0.0435	Co-occurrence
Clostridiales [F-1][G-1] bacterium HMT 093	*Bacteroidetes bacterium HMT 365*	0.4320	0.0450	Co-occurrence
V. parvula	*V. atypica*	0.4449	0.0567	Co-occurrence
V. rogosae	*Bifidobacterium*	0.4224	0.0656	Co-occurrence
Acinetobacter	*Clostridiales [F-1][G-1] bacterium HMT 093*	−0.4243	0.0673	Co-exclusion
F. fastidiosum	*Peptostreptococcaceae [XI][G-4]*	0.3254	0.0749	Co-occurrence
V. parvula	*Streptococcus oralis* subsp. *dentisani clade 398*	0.3807	0.0940	Co-occurrence
Selemonas sp. *HMT 136*	*C. durum*	0.4102	0.0943	Co-occurrence

the temporality of the observed associations and cannot prove a cause and effect relationship between submucosal microbial dysbiosis and PI. Longitudinal studies are needed to establish such relationships, but their implementation is challenging for both logistical and ethical reasons. Although our findings partly corroborate a number of earlier reports, discrepancies can conceivably be explained by differing analytic and data processing approaches, differences in sequencing platforms, length and sensitivities, and different 16s target regions (Kozich, Westcott, Baxter, Highlander, & Schloss, 2013).

Taken together, this is the first study to demonstrate an association of both previously recognized and novel taxa with the severity of human PI. Within the PI microbiome, we identified groups with significant co-occurrence and co-exclusion, suggesting that

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alterations of ecological scale, rather than single taxa, underlie the progression of a lesion. Furthermore, we demonstrated a significant positive association between the level of microbial dysbiosis and the probing depth of the PI lesion. Our findings provide a basis for further mechanistic work of the role of microbial ecological shifts and host responses in dysbiosis-associated peri-implant tissue destruction.

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CONFLICT OF INTEREST

The authors declare that there are no conflict of interests in this study.

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